

$$0 = 1 - 1 = -1 + 1 = 0$$

## From Elementary Arithmetic to Intrinsic Skein-Lasagna

Keyao Peng

IMB, Université Bourgogne Europe

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### Question

Can you prove  $0 = 0$  in a nontrivial way?

## Test your math level

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$$S^3 \sim \mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{CP}^1 \sim S^2$$



$$\pi_4(S^3) \cong \mathbb{Z}/2\mathbb{Z}$$

$$\pi_1^s(S) \cong \Omega_1^{\text{fr}}$$

$$K_1(\mathbb{F}_1) \cong S_\infty^{\text{ab}}$$

$$0=1-1=1+1=0$$

imgflip.com





## Hopf fibration

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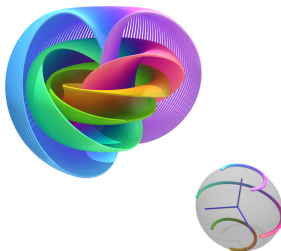
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For any point  $x \in \mathbb{S}^2$ , preimage  $H^{-1}(x)$  is a circle  $\mathbb{S}^1$ . We can visualize this map by identifying  $\mathbb{R}^3 \cong \mathbb{S}^3 - \{\infty\}$ , so that the Hopf fibration appears as a map from  $\mathbb{R}^3$  to  $\mathbb{S}^2$ :



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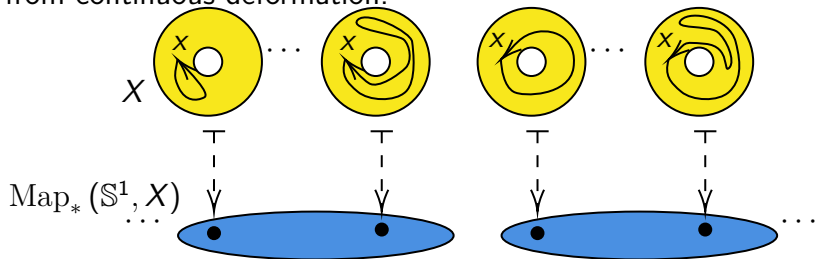
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Let us now study the case  $X = \mathbb{S}^n$ .

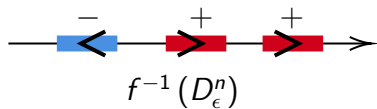
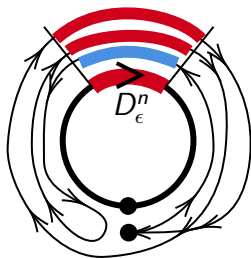


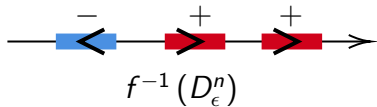
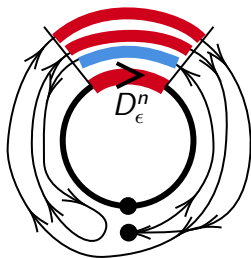
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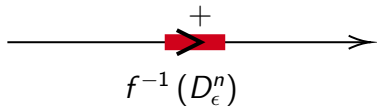
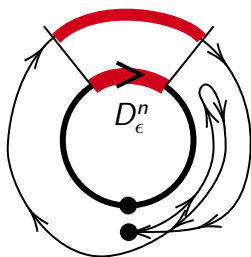
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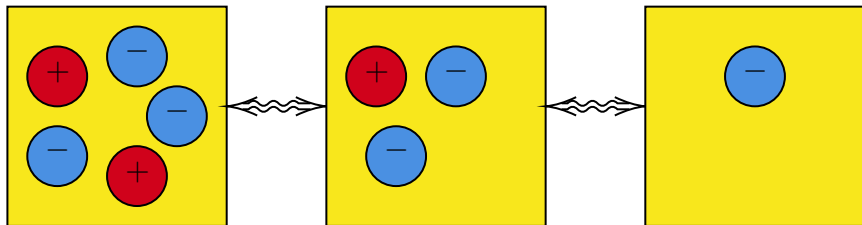
deform



## Homotopy groups of spheres

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This gives an isomorphism to the configuration space of disks.

$$\text{End}_*(\mathbb{S}^n) \cong \text{Conf}^{fr}(\mathbb{R}^n) = \bigcup_{j,k} \text{Emb}(\bigsqcup_j D_+^n \sqcup \bigsqcup_k D_-^n, \mathbb{R}^n)$$

Therefore,  $\pi_0 \text{End}_*(\mathbb{S}^n) \cong \mathbb{Z}$ ,  $(j, k) \mapsto j - k$

## Higher homotopy groups of spheres

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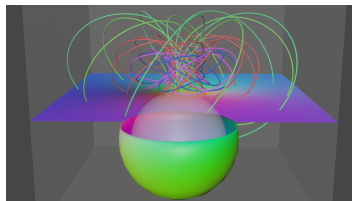
$$\text{Map}_*(\mathbb{S}^{n+1}, \mathbb{S}^n) \cong \text{Map}_*(\mathbb{S}^1, \text{End}_*(\mathbb{S}^n)) \rightarrow \text{Map}_*([0, 1], \text{End}_*(\mathbb{S}^n))$$

Then  $H$  corresponds to a path (that is, a homotopy) from  $0 \in \text{End}_*(\mathbb{S}^2)$  back to itself.

## Visualize $H$ as deformation

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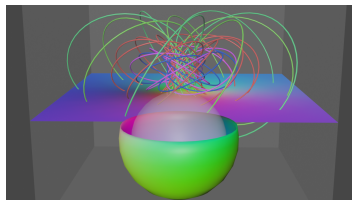
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We can also visualize it as a movement in the configuration space.  
We will show it later.

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We use the **Grothendieck group**  $K_0(\text{FinSet})$  :

$$K_0(\text{FinSet}) = \{(X, Y) \in (\text{FinSet}_{/\cong})^2\} / \{(X \sqcup Z, Y \sqcup Z) \sim (X, Y)\}$$

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But what is  $K_i(\text{FinSet})$  for  $i > 0$ ?

## Space of finite sets

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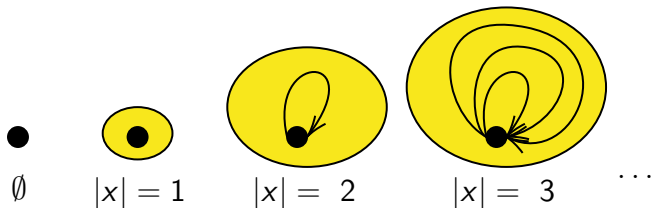
Instead of thinking the set of "finite sets", we now consider the **(moduli) space** of them  $F = \text{FinSet}^{\cong}$  :

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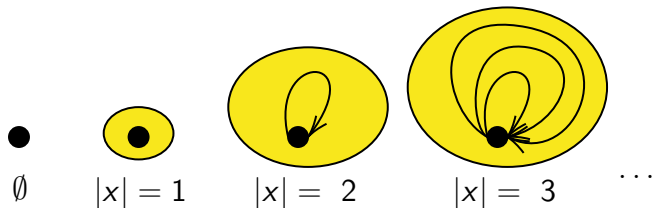
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- $\pi_0(F) = \text{FinSet}_{/\cong} \cong \mathbb{N}$ .
- $\pi_1(F, x) = S_{|x|}$ , the permutation group.
- $\pi_i(F, x) = \{0\}$  for  $i > 1$  (one can never deform an isomorphism of set to another).

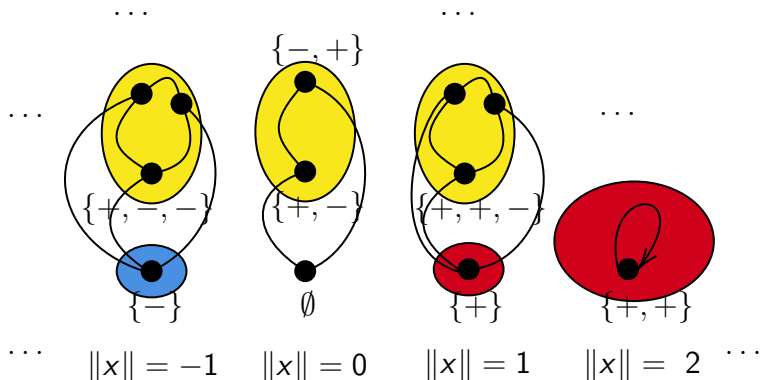
## K-space (spectrum) of finite sets

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We now want to add “negative” elements to  $F$ . To do this, we consider the space  $K(F)$  of **ordered finite sets** whose elements are marked by  $+$  or  $-$ . We identify signed sets with the same value  $\|X\| = |X_+| - |X_-|$ , and these identifications are represented by paths.

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### Answer

We can prove  $0 = 0$  nontrivially with  $h : 0 = 1 - 1 = -1 + 1 = 0!$

## Algebraic K-groups of a field

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Let  $k$  be a field. We can carry out the same construction for finite-dimensional  $k$ -vector spaces. Let  $V_k = \text{FinVect}_k^{\cong}$ , and define the **algebraic K-groups** of  $k$  by  $K_i(k) = \pi_i(K(V_k), \mathbf{0})$ . (You may think of finite sets as vector spaces over  $\mathbb{F}_1$ .)

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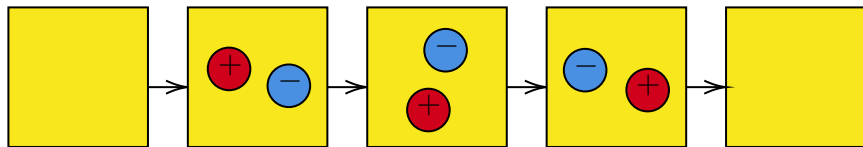
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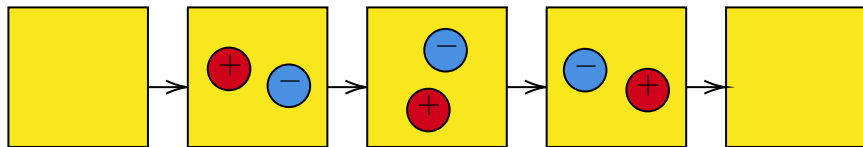
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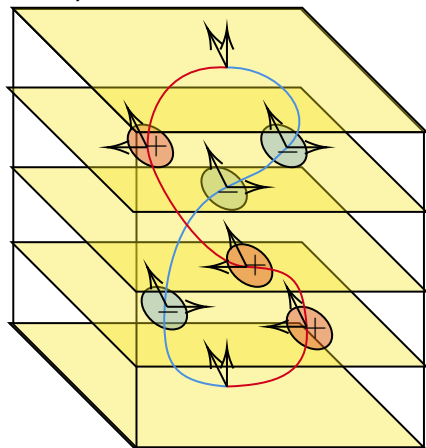
## New Question

Can we find a geometric interpretation of the general  $K$ -groups of a field  $K_i(k)$ ?

## What about cobordism?

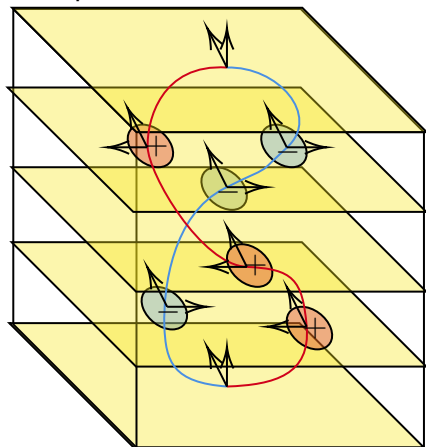
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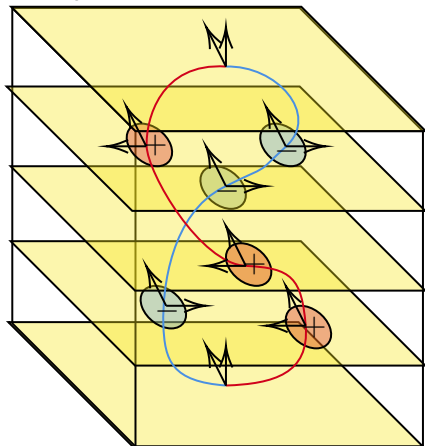
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This gives a framed circle, that is, an embedding  $\mathbb{S}^1 \hookrightarrow \mathbb{R}^n$  together with a **trivialization**  $\tau$  of the normal bundle. We write  $\Omega_i^{fr}$  for the group of cobordism classes of  $i$ -dimensional framed manifolds.

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$$\Omega_i^{fr} \cong \pi_i^s(\mathbb{S})$$

Therefore, this framed circle is nontrivial and generates  $\Omega_1^{fr}$ .

An  $n$ -**treefold** is a space that is locally homeomorphic to  $T \times \mathbb{R}^{n-1}$ , where  $T$  is a tree.

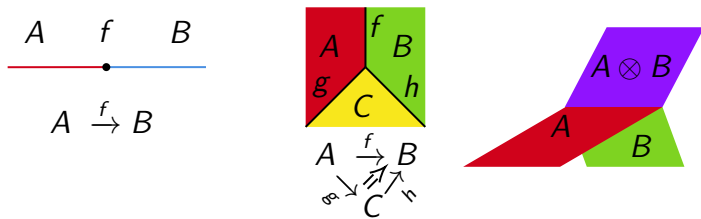


An  $n$ -**treefold** is a space that is locally homeomorphic to  $T \times \mathbb{R}^{n-1}$ , where  $T$  is a tree. Given a symmetric monoidal category  $(\mathcal{C}, \otimes)$ , define an  $n$ - $(\mathcal{C}, \otimes)$ -**lasagna**  $L$  to be a framed  $n$ -treefold  $\mathcal{T}$  with a stratification  $\mathcal{S}$ , whose strata are labeled compatibly with  $\mathcal{C}$ :

## Intrinsic Skein-Lasagna

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- 0-codimensional strata are labeled by objects; for  $i > 0$ ,  $i$ -codimensional strata are labeled by  $i$ -morphisms,
- At the branches of the treefold, we mark how pieces are combined using the monoidal product  $\otimes$ .



## Relation with "Skein-lasagna module"

We do not require an ambient space  $X$  into which the lasagna embeds (or, if you prefer, take  $X = \mathbb{R}^\infty$ ). In this sense, we study the intrinsic geometry of the lasagna itself.

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## Cobordism of lasagna

As expected, we define a cobordism relation between  $L_1$  and  $L_2$  if there exists a lasagna  $W$  with boundary such that  $\partial W = L_1 \sqcup -L_2$ . Finally, we define the  $n$ th  $(\mathcal{C}, \otimes)$ -**cobordism group**  $\Omega_n^{\mathcal{C}, \otimes}$  to be the abelian group generated by  $n$ -lasagnas modulo these cobordism relations.

## $(V_k, \oplus)$ -Cobordism Group

---

As an example, consider  $\Omega_n^{V_k, \oplus}$ , where  $V_k = \text{FinVect}_k^{\cong}$  is the category of finite-dimensional vector spaces with all isomorphisms as morphisms. In this case, one can think of a lasagna as a treefold/manifold equipped with a  **$k$ -local system**.

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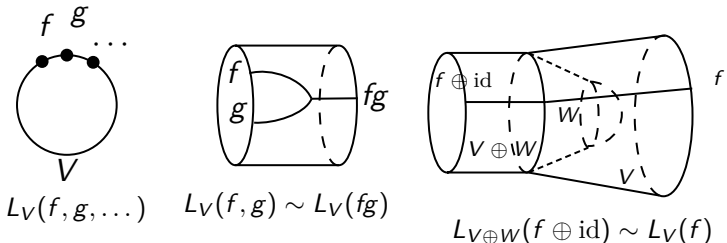
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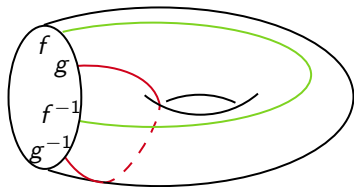
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- Clearly,  $\Omega_0^{V_k, \oplus} \cong \mathbb{Z}$ , classified by dimension of vector space.
- $\Omega_1^{V_k, \oplus}$  is generated by circles  $L_V(f)$  labeled by a vector space  $V$  and a matrix  $f \in \text{GL}_V(k)$ . By stabilizing with identity matrices, we obtain a homomorphism  $L : \text{GL}_\infty(k) \twoheadrightarrow \Omega_1^{V_k, \oplus}$ .



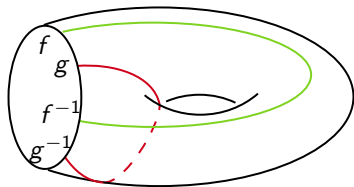


We have one more relation coming from surfaces:



$$L([f, g]) \sim 0$$

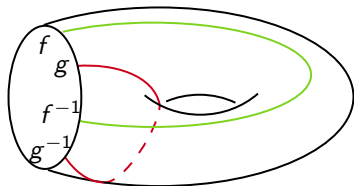
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For example, the lasagna induced by the Hopf map  $H$  is then given by  $L(-1)$ .

Recall that  $K_0(k) = \mathbb{Z}$  and  $K_1(k) = k^\times$ . This is not just a coincidence:

## Generalized Pontryagin–Thom Isomorphism

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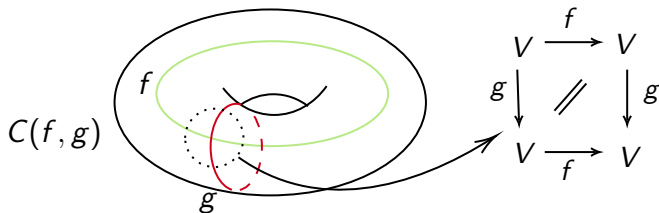
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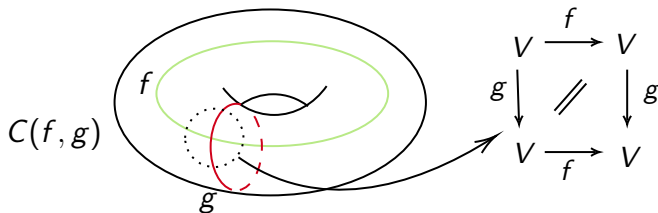
So what does this tell us when  $n = 2$ ?

For matrices  $f, g \in \mathrm{GL}(k)$  such that  $[f, g] = 1$ , one can define the following 2-lasagna  $C(f, g)$ :



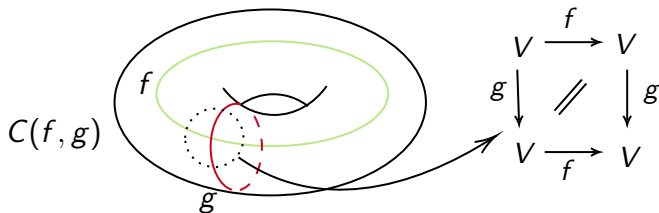


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I do not yet have Ghys' lasagna in hand (it has Heegaard genus less than 26), but I do have an easier example.

### 3-Lasagna bounds $C(e_{12}(a), e_{12}(b))$

---

For  $i \neq j$ , let the elementary matrices  $e_{ij}(a)$  be

$$(e_{ij}(a))_{kl} = \begin{cases} 1, & \text{if } k = l, \\ a, & \text{if } k = i, l = j, \\ 0, & \text{otherwise.} \end{cases}$$

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Notice that  $e_{ij}(a)^{-1} = e_{ij}(-a)$ , and these matrices satisfy the Steinberg relations:

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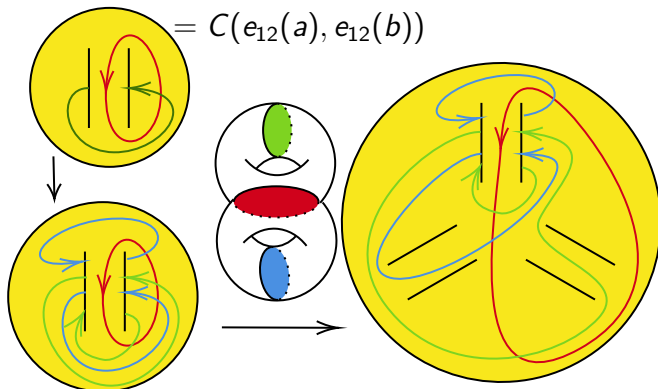
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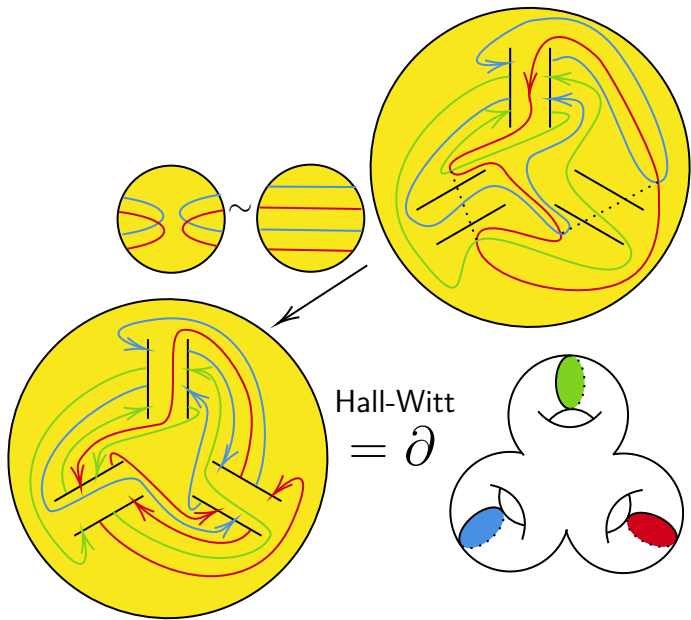
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Using these relations and the Hall–Witt identity, we obtain the following nontrivial proof that  $[e_{12}(a), e_{12}(b)] = \mathbf{1}$ :

$$\begin{aligned} [e_{12}(a), e_{12}(b)] &= [[e_{13}(a), e_{32}(1)], e_{12}(b)] \\ &= [e_{32}(1), [e_{13}(-a), e_{12}(-b)]]^{e_{13}(a)} [e_{13}(a), [e_{32}(-1), e_{12}(-b)]]^{e_{32}(-1)} \\ &= [e_{32}(1), \mathbf{1}]^{e_{13}(a)} [e_{13}(a), \mathbf{1}]^{e_{32}(-1)} = \mathbf{1} \end{aligned}$$

Following this blueprint, we obtain the following Heegaard splitting to construct a lasagna:

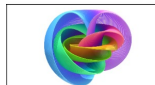




Thank you! I hope you enjoyed and gained some new insights.

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$$S^3 \sim \mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{CP}^1 \sim S^2$$



$$\pi_4(S^3) \cong \mathbb{Z}/2\mathbb{Z}$$

$$\pi_1^s(S) \cong \Omega_1^{\text{fr}}$$

$$K_1(\mathbb{F}_1) \cong S_\infty^{\text{ab}}$$

$$0=1-1=1+1=0$$

imgflip.com

